Report R10 — Permutation circuits: the same 256 rules with $V = X$

A control for the sector/attractor machinery, open boundaries

QCA Sector Analysis project (Tier 1)

2026-07-30 (`xgate_version 1x.1`, `obc0`)

Abstract

The Tier-1 machinery separates two decompositions of the computational basis: the weakly connected components (sectors, exact for both the monitored and the unmonitored dynamics) and the terminal strongly connected components (monitored attractors). In the Hadamard family those two disagree strongly, and R9 is largely about the disagreement. This report runs the identical analysis on the same 256 symbol tables with the active gate changed from the Hadamard to X. Every local symbol is then a function on bits, the transition graph has out-degree one, and three things follow at once: the two decompositions become *identical* in number, sectors and basins coincide, and the 16 rules built from `I` and `V` alone are reversible cellular automata whose sectors are exactly the permutation cycles. We verify all of this over 256 rules, confirm that the partition sum rule and the exclusion curve carry over unchanged, and record the one place where the two families genuinely differ in physics rather than in bookkeeping: the permutation circuits fragment into exponentially many, exponentially large sectors whose *attractors are almost always short*. Only 8 of 256 rules have an exponentially long cycle, 207 have a bounded one, and for the 80 V-free rules the number of exponential cycles is zero — with a complete explanation, since a V-free rule is a boolean function of its two neighbours and only the two GF(2)-affine ones sustain long orbits. Two movement analyses organise the family: the move from the sector map to the cycle map is purely vertical and is a theorem, and the move from a reversible parent to its dissipative children shows that for 177 of 240 children the resets act as the *identity* on the recurrent set, so the child's attractors are literally orbits of its parent. Two corrections to the first draft are folded in: the eight exponential orbits are now *exhibited* rather than fitted, by following a single orbit with a bitwise sweep that needs no 2^N memory — 2.1×10^9 states in one orbit at $N = 32$ for rule 57, and $N = 40$ for the rest — and the claim that the family has no dark states is withdrawn, since 127 of the 240 irreversible rules carry a decoherence-free subspace, of exponential dimension. Swept to $N = 24$ for all 256 rules. `obc0` only; `pbc` is held.

Contents

1 What changes when $V = X$

The symbol table is untouched. Each entry still says what happens at a site for one neighbour pattern; only the active gate is different:

symbol	Hadamard family (R1–R9)	this report
`I`	identity	identity
`V`	Hadamard	X, i.e. $x_i \mapsto 1 - x_i$
`D`	reset to $\|0\rangle$	reset to $\|0\rangle$
`E`	reset to $\|1\rangle$	reset to $\|1\rangle$

All four are now *functions* on the computational basis, so the circuit is one too, and $\mathrm{succ}(x)$ is a single state. Three consequences, and they are the reason this family is a useful control.

Proposition 1 (functional graph)**.** *The one-cycle transition graph has out-degree exactly* 1.

Theorem 1 (F1: one cycle per component)**.** *In a functional graph every weakly connected component contains exactly one cycle. Hence for every X-gate rule, every N and every boundary condition,*

$$n_{\mathrm{rec}} = n_{\mathrm{wcc}}, \tag{1}$$

the basin of each cycle is its entire component, and no state lies in two basins.

That identity is exactly what fails in the Hadamard family, and fails badly: R9's rule 22 at $N = 12$ has three monitored attractors, each a single state, sitting inside two sectors that cover all 4096 states. There A4 could only be asserted in one direction. Here the two tiers coincide by construction, so any disagreement in the code would show up immediately — which makes this family a test of the machinery as much as a physics question.

1.1 What "reversible" means here, and what it does not

The word is overloaded in this area, so it is fixed explicitly.

Definition 1. A rule is *reversible at* $(N, \texttt{bc})$ if its one-cycle map $\{0,1\}^N \to \{0,1\}^N$ is a bijection.

Proposition 2. *A rule is reversible, for every* $N \geq 3$ *and both boundary conventions, iff its symbol table uses only* `I` *and* `V`*. There are* $2^4 = 16$ *such rules:* 51*,* 54*,* 57*,* 60*,* 99*,* 102*,* 105*,* 108*,* 147*,* 150*,* 153*,* 156*,* 195*,* 198*,* 201*,* 204*.*

Proof sketch. One elementary step reads the two neighbour bits of site i and then applies `I` or X to site i. The controls are bits $\neq i$ and are untouched by that step, so the step is a controlled-X and hence a bijection; a composition of bijections is a bijection, whatever the layer order. Conversely `D` and `E` send both values of the target bit to the same value, so two states differing only in site i but sharing its neighbour pattern collide. □

Note the proof does not use the sublattice structure, so it survives the case where the even sites fail to be an independent set (`pbc` at odd N). Checked exhaustively anyway at $N = 7, 8, 9$ for both conventions: exactly those 16 are bijections and none of the other 240 is.

Not the six reversible ECA. A better-known count lives nearby and the two must not be conflated. For *elementary* cellular automata — one layer, simultaneous update, each cell a function of its three-cell neighbourhood — only **six** of the 256 rules are reversible: 15, 51, 85, 170, 204, 240, the shifts and complements.

Two things differ, not one. The construction differs: the brick-wall QCA updates one sublattice at a time and applies a gate to the target conditioned on its neighbours, which makes invertibility a purely *local* property of the gate. And the quantifier differs: the ECA six are those bijective for *every* N, an intersection over lattice sizes, and at a fixed N the bijective set is strictly larger — on a ring of $N = 7$ it also contains 45, 75, 89 and 150. Our 16 need no such intersection: the proposition above is proved locally and holds at every N separately, which is why the two counts are not comparable term by term. Only 51 and 204 lie in both sets.

They are ECAs, of the same number. The complementary fact is worth recording, because it is what connects this family to the literature. For a rule built from `I` and `V` the local update is

$$x_i \longmapsto x_i \oplus f(x_{i-1}, x_{i+1}),$$

with f the indicator of `V` in the symbol table, i.e. one of the 16 boolean functions of two arguments. That map is an elementary cellular automaton, and — checked for all 16 — it is the ECA of *the same Wolfram number* as the rule. So the reversible family is exactly the set of ECAs that are affine in the centre cell, applied brick-wall; being affine in the centre cell is what makes the brick-wall version a bijection, which is the proposition above in another language. In particular X_{54} (`IVVV`) is $x_i \mapsto x_i \oplus (x_{i-1} \vee x_{i+1})$, the Rule 54 automaton, which is the most-studied exactly solvable QCA in the literature; R11 §7.5 develops that connection.
So the 16 rules that were "unitary" in the Hadamard family are here *reversible* in the sense just defined: every state is recurrent, each component is a single cycle, and the sector decomposition *is* the orbit decomposition of a permutation of $\{0,1\}^N$.

Implementation. The site order and the neighbour conventions are not reimplemented. The compiled step list comes straight from `core/cycle.py::_compile`, so the brick-wall order (even sites ascending, then odd, each reading the current state) and the boundary handling are byte-for-byte those of the validated engine; only the local action is swapped. Nothing in the Hadamard engine is touched, and the two live in separate stores. This matters because the brick-wall order is *not* two simultaneous sublattice updates once the even sites stop being an independent set.

Cost. The whole analysis is one $O(2^N)$ pass over a functional graph — a three-colour walk that finds the cycles and, on the way back, each node's distance to one. No amplitudes, no Tarjan stack, no union-find needed (though we run union-find anyway as a cross-check, §12). All 256 rules at $N = 24$ — sixteen million states each — take 43 min in total, against hours per *rule* for the Hadamard sweep at $N = 16$. The map itself is built with numpy, lifting the loop over x out of Python while keeping the site order from `_compile`, which is what makes $N = 24$ affordable; the vectorised and scalar builders are checked against each other in the tests.

2 Coverage

N	rules covered	2^N
6	256	64
7	256	128
8	256	256
9	256	512
10	256	1 024
11	256	2 048
12	256	4 096
13	256	8 192
14	256	16 384
15	256	32 768
16	256	65 536
17	256	131 072
18	256	262 144
19	256	524 288
20	256	1 048 576
21	256	2 097 152
22	256	4 194 304
23	256	8 388 608
24	256	16 777 216

Uniform over all 256 rules. `pbc` is deliberately not run: the ring has its own caveats and is held for a separate report.

3 The headline: large sectors, short attractors

family	rules	exp. sectors	exp. cycles	median cyclic frac.	median depth
reversible	16	6	6	1	0
V+reset	160	155	2	1.79×10^{-7}	13
V-free	80	78	0	1.46×10^{-5}	12

Read the two middle columns against each other. 239 **of** 256 **rules have exponentially large sectors, and only** 8 **have exponentially long cycles.** For the 80 V-free rules the count is *zero*: not one of them has a cycle whose length grows exponentially, while 78 of them have exponentially large sectors. The median cyclic fraction — the share of the basis that is periodic at all — is 1.8×10^{-7} for the V+reset family and 1.5×10^{-5} for the V-free family at $N = 24$, with a median transient depth of 13 and 12 steps.

The full cycle-growth census refines the middle column, and is the table to read if one wants the shape of the collapse rather than its headline:

family	rules	bounded	linear	exponential	irregular
reversible	16	6	4	6	0
V+reset	160	123	23	2	12
V-free	80	78	0	0	2
all	256	207	27	8	14

207 of 256 rules have a cycle length that does not grow with N at all, 27 grow *linearly*, 8 exponentially and 14 are irregular. Of the irregular ones all but two simply alternate with the parity of N (rule 23 runs $21, 1, 23, 1, 25$ over $N = 20 \dots 24$: at odd N its longest cycle is a single fixed point); the two that are genuinely erratic are 90 and 165, and §7 says why.

A correction to the first draft of this report. That count was 24 in the first version, and 16 of the 24 were fitting artefacts. Two failure modes, both invisible to the sector series and both exposed by the cycle series, have been fixed in `scaling/sectors.py`:

- Over a window of a dozen sizes $\ln N$ is very nearly an affine function of N, so the volume-fraction discriminator — which decides whether $D_{\max} = c\,2^N N^\alpha$ by asking whether $D_{\max}/2^N$ is better explained by a power of N than by an exponential — could be won by the power-law model on a *bounded* series, whose ratio falls like $-N \ln 2$. Rule 45's longest cycle is $1, 2, 2, 1, 2, 2, 1, 2$ and was reported as growing like 2^N. The rule now requires the series to be eventually non-decreasing and $|\alpha|$ to be a plausible prefactor power (≤ 8; the steepest genuine sector case in R9 is -4.61).
- BIC selects the three-parameter model M2 for a straight *line* as well, because $\ln y$ is concave: the growth goes into $\alpha \ln N$ and the leftover κ is small but not zero, so the class came out "exponential" and the two-parameter fit then reported a base near 1.1 for an arithmetic progression. Rule 1's cycle lengths run $44, 47, 50, \dots, 65$, exactly $+3$ per site. The leave-one-out band of M2's κ settles it — for rule 1 it lies entirely *below* 1 — so an exponential label now requires that band to exclude zero rate.

Neither guard touches a single one of R9's 512 sector descriptors (checked by rebuilding the map and diffing: zero changes, identical verdict tally), which is the expected outcome — these are cycle-series pathologies. The number of rules with exponentially large *sectors* is likewise unchanged at 239. Every claim below uses the corrected classification.

So the permutation circuits are the textbook classical picture: a state falls through a long transient into a short attractor, and essentially all of the Hilbert space is transient. The sector structure is large and the recurrent structure is tiny, and the gap between them is the whole content of the family.

3.1 The eight rules above the curve in the cycle map

Nothing forces a point up there. In the sector panel $ab \geq 2$ is an exclusion curve and every rule must respect it; in the cycle panel it is only a drawn reference, so a rule with $a\,b_{\text{rec}} > 2$ is making a genuine claim — it has exponentially many attractors *and* an exponentially long one. Exactly eight rules do, and they are the eight of §7's exponential-cycle list:

rules	tuple	family	a	b_{rec}	ab
$57, 99$	`VIVV`, `VVIV`	reversible	1.4399	2.0000	2.8798
$156, 198$	`IIVI`, `IVII`	reversible	2.0000	1.2744	2.5488
201	`VIII`	reversible	1.8580	1.2593	2.3398
108	`IIIV`	reversible	1.8430	1.2679	2.3367
73	`VIID`	V+reset	1.6536	1.2592	2.0822
109	`EIIV`	V+reset	1.6398	1.2679	2.0792

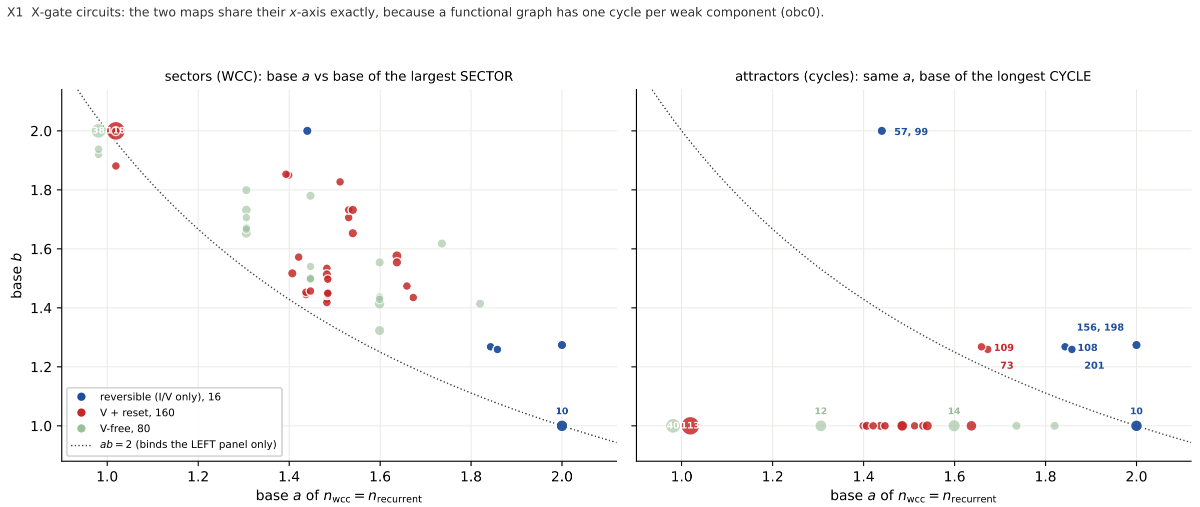

Figure 1: **X1, the two maps.** Left: base of the sector count against base of the largest *sector*. Right: the same x-axis — identical by (1), not merely similar — against the base of the longest *cycle*. The dotted curve is $ab = 2$; it binds the left panel, because sectors partition, and is only a reference on the right, because cycles do not. The collapse of the right panel onto $b = 1$ is the headline: marker area grows with the number of rules in a cell and the two largest cells there hold 113 and 40 rules. The eight rules that nonetheless lie *above* the curve on the right are named there and tabulated in §3.1; the ten sitting exactly *on* it are all at $(2, 1)$ and are listed in the text.

Six are reversible, where a long orbit costs nothing, and the two irreversible ones are 73 and 109 — the two children that inherit an exponential cycle from a parent that has one (§8). Both sit just above the curve, at 2.08, because the reset has thinned the attractor count without touching the longest orbit.

The sequences, so the claim can be checked rather than taken on the fitted bases:

rules	series	$N = 19 \dots 24$
$57, 99$	n_{rec}	242, 362, 524, 792, 1 196, 1 798
	$L_{\max}$	244 632, 489 856, 925 416, 2 025 296, 3 735 008, 6 807 384
$156, 198$	n_{rec}	50 484, 93 142, 176 458, 328 728, 624 312, 1 170 866
	$L_{\max}$	140, 210, 210, 420, 420, 420
201	n_{rec}	78 250, 146 460, 274 180, 514 398, 964 552, 1 812 802
	$L_{\max}$	238, 253, 340, 391, 462, 561
108	n_{rec}	105 382, 195 970, 364 808, 680 106, 1 269 584, 2 373 460
	$L_{\max}$	154, 187, 238, 253, 340, 391
73	n_{rec}	8 426, 14 327, 23 960, 40 490, 67 029, 114 189
	$L_{\max}$	238, 253, 340, 391, 462, 561
109	n_{rec}	7 866, 13 159, 21 925, 36 411, 61 040, 102 737
	$L_{\max}$	154, 187, 238, 253, 340, 391

Read them in pairs. 57 and 99 are the extreme case: only 1798 orbits at $N = 24$ but the longest runs through 6 807 384 of the 16 777 216 states, i.e. 41% of the space in a single cycle — $b_{\text{rec}} = 2$ with $a = 1.44$, the highest product in the family. 156 and 198 are the opposite arrangement at almost the same product: 1.17 million orbits, none longer than 420, with the length climbing through the highly composite values $140, 210, 420$ in the manner of an lcm rather than a growth law. 201 and 73 share their cycle series exactly $(238, 253, \dots, 561)$, as do 108 and 109 $(154, 187, \dots, 391)$ — that is the inheritance of §8 visible in the raw numbers, the child keeping the parent's longest orbit while its own attractor count is an order of magnitude smaller (114 189 against 1 812 802; 102 737 against 2 373 460).

Ten further rules sit exactly *on* $ab = 2$, all in the single cell $(2,1)$: 51, 54, 60, 102, 105, 147, 150, 153, 195, 204. All ten are reversible, so every state is periodic and the sector count is forced to $\Theta(2^N/L_{\max})$ with $L_{\max}$ bounded or linear — 204 has 2^N fixed points, 51 has 2^{N-1} two-cycles, 60, 102, 153 and 195 have cycle length 32 at every $N \geq 19$, and 105, 150, 54, 147 grow linearly. They are on the curve for the same bookkeeping reason the identity is on it in the sector map, not because anything constrains them.

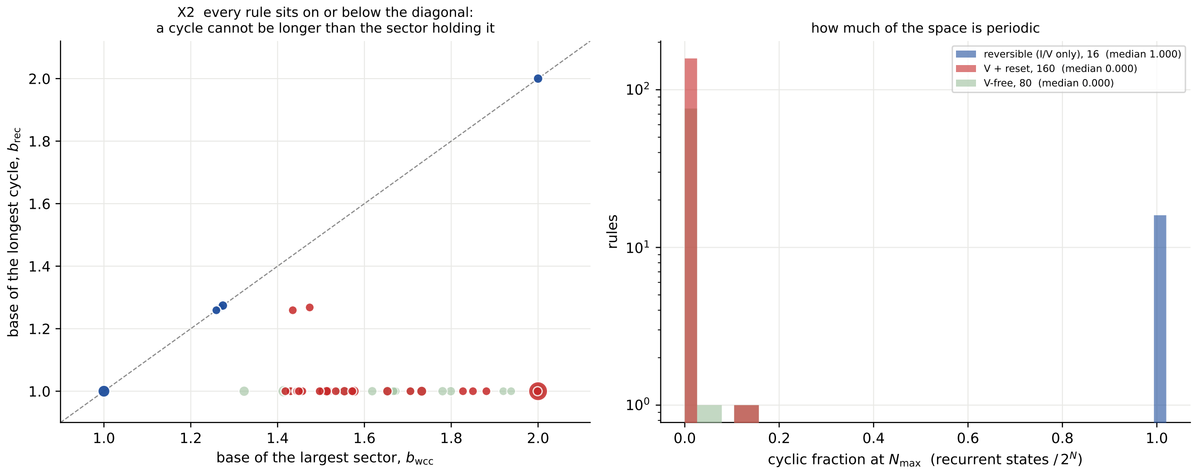

Figure 2: **X2, the drop, per rule.** Left: longest cycle against largest sector. Every rule lies on or below the diagonal — a cycle cannot be longer than the sector containing it — and the reversible rules are exactly the ones *on* it, since for them the sector *is* the cycle. Right: the distribution of the cyclic fraction, log-scaled, by family.

4 The reversible rules

For the 16 rules built from `I` and `V` the two decompositions are not just equal in number: the sectors *are* the cycles, elementwise. So a single table describes both maps.

rule	tuple	a	b	ab	n_{wcc}	$D_{\max}$
51	`VVVV`	2.00000	1.00000	2.0000	8 388 608	2
54	`IVVV`	2.00000	1.00000	2.0000	364 724	67
57	`VIVV`	1.43992	2.00000	2.8798	1 798	6 807 384
60	`IIVV`	2.00000	1.00000	2.0000	526 356	32
99	`VVIV`	1.43992	2.00000	2.8798	1 798	6 807 384
102	`IVIV`	2.00000	1.00000	2.0000	526 356	32
105	`VIIV`	2.00000	1.00000	2.0000	671 092	25
108	`IIIV`	1.84296	1.26792	2.3367	2 373 460	391
147	`VVVI`	2.00000	1.00000	2.0000	364 724	46
150	`IVVI`	2.00000	1.00000	2.0000	671 092	25
153	`VIVI`	2.00000	1.00000	2.0000	524 288	32
156	`IIVI`	2.00000	1.27441	2.5488	1 170 866	420
195	`VVII`	2.00000	1.00000	2.0000	524 288	32
198	`IVII`	2.00000	1.27441	2.5488	1 170 866	420
201	`VIII`	1.85800	1.25932	2.3398	1 812 802	561
204	`IIII`	2.00000	1.00000	2.0000	16 777 216	1

Three anchors are exact and worth stating, because they are the analogues of R9's:

- **Rule** 204 (`IIII`) is the identity: 2^N fixed points, $a = 2$, $b = 1$, $ab = 2$ — on the curve, as in the Hadamard case, but for a different reason (there every sector was a frozen singleton; here every *cycle* is).
- **Rule** 51 (`VVVV`) is the global flip $x \mapsto \bar{x}$: 2^{N-1} two-cycles, so $a = 2$, $b = 1$ and $ab = 2$. Note the contrast with the Hadamard family, where 51 gives *one* sector of size 2^N — the same symbol table, opposite extremes of the sector axis, purely from the gate.
- **Rule** 150 (`IVVI`) with $V = X$ is the linear map $x_i \mapsto x_i \oplus x_{i-1} \oplus x_{i+1}$ applied brick-wall. Its longest cycle is short and grows *linearly*: $D_{\max} = 14, 15, 16, 17, 18$ at $N = 13 \ldots 17$, i.e. $N+1$ over the range computed, with $n_{\text{wcc}} \approx 2^N/(N+1)$ so $a = 2$, $b = 1$, $ab = 2$. Against the Hadamard rule 150, whose sectors are the binomial domain-wall shells of R8, this is a completely different object built from the same table.

The family spans the whole plane. At one end 204 and 51 have 2^N and 2^{N-1} tiny orbits; at the other, 57 and 99 (`VIVV`, `VVIV`) have only 110 sectors at $N = 17$ with the largest holding 36 456 states, giving $a = 1.405$, $b = 1.961$ and $ab = 2.755$. Four rules — 60, 102, 153, 195 — and also 108 and 201 sit at $b = 2$ exactly with $a \approx 1.79$–1.85, so $ab \approx 3.6$–3.7, far above the exclusion curve: many orbits *and* a largest orbit that is a fixed fraction of the space.

5 Sectors, recurrent sets and cycles are three different sizes

The rest of this report is about a question that the two panels of Figure 1 pose immediately: how can 239 rules have exponentially large *sectors* while only 8 have exponentially long *cycles*? The answer is that these count different things, and only one of them is obliged to be large.

Three quantities. For a functional graph, write

$$\text{sectors } \{S_k\}, \qquad \text{Rec} = \bigcup_k C_k \text{ (the union of the cycles)}, \qquad L_k = |C_k|.$$

Then $\sum_k |S_k| = 2^N$ exactly — sectors partition the basis — while $\sum_k L_k = |\text{Rec}|$, which is whatever the dynamics leaves behind. At $N = 24$ the median irreversible rule has $|\text{Rec}|/2^N \approx 3.6 \times 10^{-7}$: six states out of sixteen million.

Only the sector map has an exclusion curve. The hyperbola is a statement about a partition and nothing else. With $n_{\text{wcc}} \sim a^N$ and $D_{\max} \sim b^N$,

$$a^N b^N \;\gtrsim\; \sum_k |S_k| \;=\; 2^N \quad \Longrightarrow \quad ab \geq 2.$$

Run the same argument on the cycles and it gives $n_{\text{rec}}\, L_{\max} \geq |\text{Rec}|$, i.e. $a\, b_{\text{rec}} \geq c$ where c is the growth base of the recurrent set — *not* 2. And here c is essentially a itself: almost every cycle is a fixed point, so $|\text{Rec}| \approx n_{\text{rec}}$, and fitting both series gives $|c - a| < 0.01$ for 222 of the 240 rules where both bases exist. The constraint on the cycle map therefore degenerates to $b_{\text{rec}} \geq 1$, which is no constraint at all. **The sector map is forced to be exponential somewhere; the cycle map is free to collapse, and does.** That is the whole content of the two panels, and it is bookkeeping, not physics.

The physics underneath is an anticorrelation. A large sector is evidence of information *loss*: a component is big because many states share a future, i.e. because the preimage trees are bushy. A long cycle is evidence of information *preservation*: to return to x after L steps the map must be injective all along the orbit. Contraction inflates the first and deflates the second *simultaneously*, so the two coordinates are not merely uncorrelated, they are pushed in opposite directions. Only a bijection can be large in both, which is precisely why the reversible rules are the ones on the diagonal of Figure 2 and the only family with a full column of exponential cycles.

How unusual is that? A uniformly random map on $n = 2^{24}$ points has, by the classical Flajolet–Odlyzko estimates, about $\sqrt{\pi n/2} \approx 5100$ cyclic points, a longest cycle near $0.78\sqrt{n} \approx 3200$, a tail length $\sqrt{\pi n/8} \approx 2600$ and only $\frac{1}{2}\ln n \approx 8$ components. The median irreversible X-gate rule has **six** cyclic points and a transient depth of 13: three orders of magnitude fewer recurrent states and two orders of magnitude shallower than a random map of the same size. A random map would have exponentially long cycles, with base $\sqrt{2}$. So the collapse of the cycle map is not a generic property of irreversible maps — these circuits are far more strongly contracting than chance, which is what locality plus a reset buys.

An exponentially large recurrent set with only linear cycles. The three sizes really are independent. Rule 1 (`VDDD`) has a recurrent set of size F_{N+2} — Fibonacci, base φ — and it is not merely the count that matches: the recurrent set *is* the golden-mean shift,

$$\mathrm{Rec}(X_1) \;=\; \{x \in \{0,1\}^N : x \text{ contains no } 11\},$$

verified by set equality at $N = 10, 12, 13$ and by cardinality over the whole sweep. Its longest cycle meanwhile is $44, 47, 50, \ldots, 65$, growing by exactly 3 per site. Fifteen further rules share the count F_{N+2} and six more the shifted counts F_{N+1} and F_N. So one can have an exponentially large attractor set made almost entirely of *fixed points*: exponential degeneracy without any long-time dynamics — the classical analogue of a large ground-space rather than of a long orbit. That distinction is invisible in the base plane, which is why the growth-class table above is the honest presentation.

6 Movement 1: from the sector map to the cycle map

The two panels of Figure 1 are the same 256 rules seen twice, so the natural object is the *move*, not the pair of pictures. Drawing it in one plane makes an easy theorem visible:

Proposition 3. *For every X-gate rule the move from the sector map to the cycle map is a straight vertical drop: the horizontal coordinate is identical by (1), and $L_k \le |S_k|$ pointwise, so $b_{\mathrm{rec}} \le b_{\mathrm{wcc}}$ always.*

In the Hadamard family (R9's F3) the corresponding move had both components, and the horizontal one — sectors merging into shared basins — carried the interest. Here that component is zero by construction, and what is left is a clean measurement of how much of a basin is periodic.

family	rules with both bases	no move	median drop	max drop	median cyclic frac.
reversible	16	16	0.000	0.000	1
V+reset	147	0	1.000	1.000	1.19×10^{-7}
V-free	78	0	0.929	1.000	1.49×10^{-6}

The distribution is close to two-valued. The 16 reversible rules do not move at all — for them the sector *is* the cycle, so the drop is exactly 0 and this is a check on the pipeline, not a measurement. Every other rule moves, with a median drop of exactly 1.000: the modal irreversible rule has an exponentially large sector, $b_{\mathrm{wcc}} = 2$, and a bounded cycle, $b_{\mathrm{rec}} = 1$. Nothing lands *above* the diagonal, as the proposition requires (0 of 241).

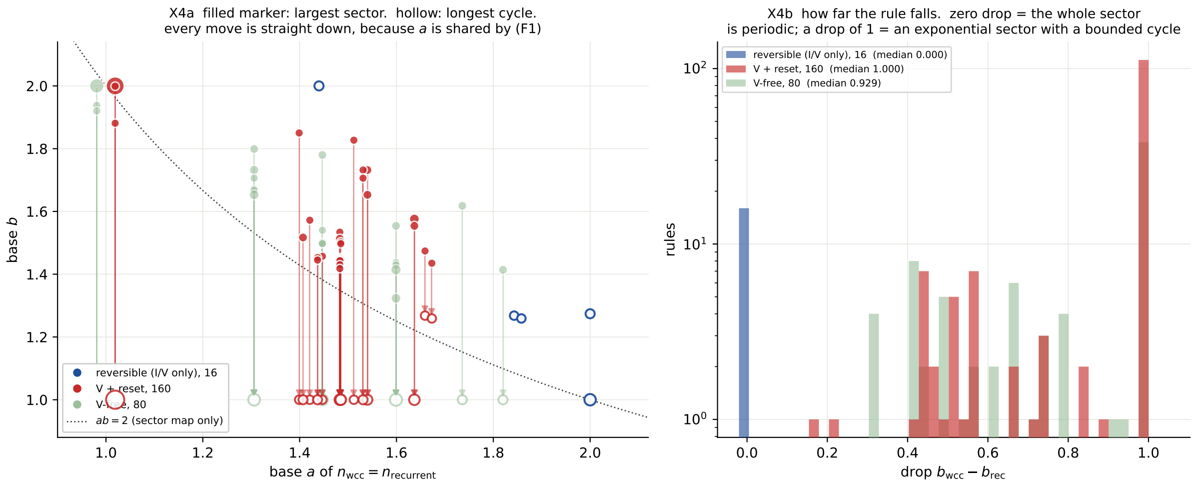

Figure 3: **X4, movement WCC → recurrent.** Left: filled marker at the largest sector, hollow marker at the longest cycle, joined by the move. Every arrow is vertical — that is (1), not an artefact of the layout — and every arrow points down. The reversible rules (blue) have no arrow. Right: the drop, log-scaled counts; the pile at 1.0 is the modal rule, exponential sector and bounded cycle.

7 Why the V-free family collapses

The sharpest case of the collapse is the V-free family: 78 of 80 have exponentially large sectors and *none* has an exponentially long cycle. Here the mechanism is completely identifiable rather than statistical.

A V-free rule's symbols are `I` (keep x_i), `D` (set 0) and `E` (set 1). Take the extreme case first: if *all four* symbols are resets — 16 rules — then the new bit does not depend on the old bit at all, and the rule is literally a boolean function of the two neighbours, $x_i \mapsto f(l, r)$. There are exactly 16 such functions and the rules enumerate them:

rule	tuple	$x_i \mapsto f(l,r)$	GF(2)-affine	cycle class	longest cycle seen
0	`DDDD`	0	no	constant	1
5	`EDDD`	$\neg(l \vee r)$	no	constant	1
10	`DEDD`	$\neg l \wedge r$	no	constant	1
15	`EEDD`	$\neg l$	yes	constant	1
80	`DDED`	$l \wedge \neg r$	no	constant	1
85	`EDED`	$\neg r$	yes	constant	1
90	`DEED`	$l \oplus r$	yes	irregular	2 047
95	`EEED`	$\neg(l \wedge r)$	no	constant	1
160	`DDDE`	$l \wedge r$	no	constant	1
165	`EDDE`	$\neg(l \oplus r)$	yes	irregular	2 047
170	`DEDE`	r	yes	constant	1
175	`EEDE`	$\neg l \vee r$	no	constant	2
240	`DDEE`	l	yes	constant	1
245	`EDEE`	$l \vee \neg r$	no	constant	2
250	`DEEE`	$l \vee r$	no	constant	1
255	`EEEE`	1	no	constant	1

Fourteen of the sixteen are constant, a (possibly negated) projection, or monotone, and every one of them has a longest cycle of 1 or 2 at every size computed. The two that are neither are $f = l \oplus r$ and its negation: rules 90 (`DEED`) and 165 (`EDDE`). Those two are GF(2)-affine, so the sweep is a linear map over GF(2), its orbit lengths are multiplicative orders, and they are exactly the two V-free rules in the whole family whose cycle length is not bounded: $63, 62, 2047, 8, 1023$ at $N = 20 \ldots 24$, erratic in the way orders of matrices over a finite field are erratic. *The XOR column is the entire exception.*

The remaining 64 V-free rules mix `I` with resets, and all 64 have a bounded cycle. So across the family the pattern is: with no X available, the only way to keep a bit oscillating is to compute it linearly from its neighbours, and only two of the sixteen local functions do that.

Contrast the rules whose cycles *do* grow:

rule	tuple	class	base	parent	longest cycle, $N = 20 \ldots 24$
57	`VIVV`	exponential	2.0000	57	489 856, 925 416, 2 025 296, 3 735 008, 6 807 384
99	`VVIV`	exponential	2.0000	99	489 856, 925 416, 2 025 296, 3 735 008, 6 807 384
156	`IIVI`	exponential	1.2744	156	210, 210, 420, 420, 420
198	`IVII`	exponential	1.2744	198	210, 210, 420, 420, 420
108	`IIIV`	exponential	1.2679	108	187, 238, 253, 340, 391
109	`EIIV`	exponential	1.2679	108	187, 238, 253, 340, 391
201	`VIII`	exponential	1.2593	201	253, 340, 391, 462, 561
73	`VIID`	exponential	1.2592	201	253, 340, 391, 462, 561
23	`EVVD`	irregular	−−	150	21, 1, 23, 1, 25
33	`VDDV`	irregular	−−	105	21, 2, 23, 2, 25
41	`VIDV`	irregular	−−	105	21, 2, 23, 2, 25
90	`DEED`	irregular	−−	204	63, 62, 2 047, 8, 1 023
97	`VDIV`	irregular	−−	105	21, 2, 23, 2, 25
107	`VEIV`	irregular	−−	105	3, 8, 3, 8, 3
121	`VIEV`	irregular	−−	105	3, 8, 3, 8, 3
122	`DEEV`	irregular	−−	108	19, 2, 21, 2, 23
123	`VEEV`	irregular	−−	105	19, 2, 21, 2, 23
161	`VDDE`	irregular	−−	201	21, 2, 23, 2, 25
165	`EDDE`	irregular	−−	204	63, 62, 2 047, 8, 1 023
169	`VIDE`	irregular	−−	201	21, 2, 23, 2, 25
225	`VDIE`	irregular	−−	201	21, 2, 23, 2, 25
233	`VIIE`	irregular	−−	201	21, 2, 23, 2, 25

Every single one contains a `V`, apart from 90 and 165. This is the answer to the question the family poses: exponential sector structure is forced by the sum rule and is available to any strongly contracting map, whereas an exponentially long cycle needs a genuine local *flip*, and 80 rules do not have one. Note also that most of the "irregular" cycle rules are simply parity-alternating — rule 23 runs $21, 1, 23, 1, 25$, its longest cycle collapsing to a fixed point at odd N — so the truly erratic set is $\{90, 165\}$ alone.

8 Movement 2: from a reversible parent to its children

Every rule has a coherent parent: switch the resets off, $\mathtt{D}, \mathtt{E} \to \mathtt{I}$ (`rules.coherent_part`). The parent is always one of the 16 reversible rules, and a parent with k identity slots has $3^k - 1$ children, so the 16 clusters cover all 240 irreversible rules — the identity 204 alone accounts for the whole V-free family of 80. This is R9's F6 clustering carried over, and here it can be pushed further than R9 could, because the parent's dynamics is a permutation and one may ask whether a child's attractor is literally an *orbit of its parent*.

parent	tuple	children	parent cycle	exp. sectors	exp. cycles	inherit	median drop
54	IVVV	2	polynomial	0	0	2/2	1.000
57	VIVV	2	exponential	2	0	0/2	0.576
60	IIVV	8	constant	0	0	4/8	1.000
99	VVIV	2	exponential	2	0	0/2	0.576
102	IVIV	8	constant	0	0	4/8	1.000
105	VIIV	8	polynomial	2	0	2/8	0.554
108	IIIV	26	exponential	11	1	20/26	1.000
147	VVVI	2	polynomial	0	0	0/2	1.000
150	IVVI	8	polynomial	1	0	7/8	1.000
153	VIVI	8	constant	0	0	0/8	1.000
156	IIVI	26	exponential	3	0	23/26	1.000
195	VVII	8	constant	0	0	0/8	1.000
198	IVII	26	exponential	3	0	23/26	1.000
201	VIII	26	exponential	11	1	18/26	1.000
204	IIII	80	constant	38	0	74/80	0.929

8.1 Inheritance: the reset acts as the identity on the recurrent set

Proposition 4 (verified, not proved)**.** *For* 177 *of the* 240 *children, the child's map and its parent's map agree on the child's recurrent set. Consequently every attractor of such a child is an orbit of its reversible parent: dissipation* prunes *the parent's orbit list without creating new periodic behaviour.*

Checked exhaustively at $N = 12$ and $N = 13$ — one size of each parity, since the brick-wall sublattices are only independent sets at even N — with the same 177 rules passing at both. The 63 exceptions do manufacture attractors of their own, but with the two GF(2)-affine rules of §7 set aside they are all short: at $N = 12$ the longest among them is a 13-cycle, against 63 for 90 and 165. And the ordering $L_{\max}(\text{child}) \leq L_{\max}(\text{parent})$ holds for 234 of 240; the six exceptions (90, 165, 173, 175, 229, 245) are all descendants of the identity 204, whose own longest cycle is 1, so the ceiling they exceed is the degenerate one.

This is the cleanest statement in the report about what dissipation does to recurrence, and it explains the second column of the parent table. An exponentially long cycle can only be *inherited*, so it can only appear below a parent that has one; six reversible rules do (57, 99, 108, 156, 198, 201), and exactly two children manage to keep it:

- 73 (`VIID`) under 201 (`VIII`): identical longest-cycle series, $253, 340, 391, 462, 561$ at $N = 20 \ldots 24$. Every cycle of 73 at $N = 12$ is a cycle of 201 — all 253 of them — inside a recurrent set that the reset has cut down from 1018 orbits to 253.
- 109 (`EIIV`) under 108 (`IIIV`): likewise, $187, 238, 253, 340, 391$, and 243 of 108's 1442 orbits survive at $N = 12$.

Both are minimally dissipative — one reset symbol in the table — and both keep the parent's *longest* orbit, not merely a long one. The cyclic fraction, however, is destroyed: 0.15 and 0.11 against the parent's 1. The attractor survives; the basin structure around it does not.

9 What carries over, and what does not

Carries over. The sum rule and the exclusion curve are properties of a *partition*, and sectors partition here exactly as they did there. The fit-free finite-N form

$$n_{\text{wcc}}(N) \cdot D_{\max}(N) \;\geq\; 2^N$$

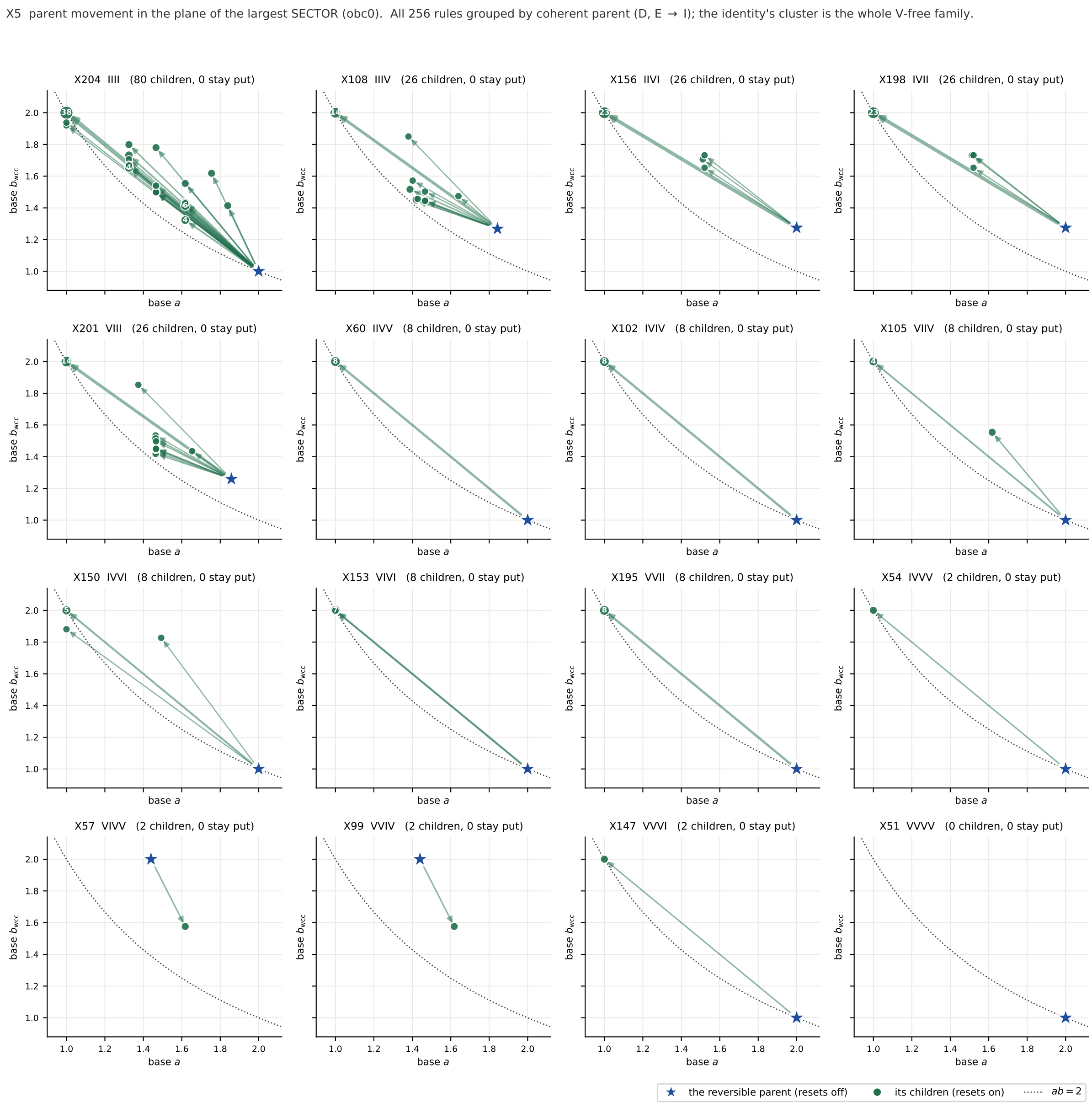

Figure 4: **X5, parent movement in the sector plane.** Star: the reversible parent. Dots: its children. 227 of the 233 arrows point up *and* to the left — adding a reset merges sectors (fewer of them, so a falls) and inflates the survivors (so b rises), sliding the cluster along and above the exclusion curve. The visible exceptions are the clusters of 57 and 99, whose children move *down* from $b = 2$, and one child of 108 that moves right.

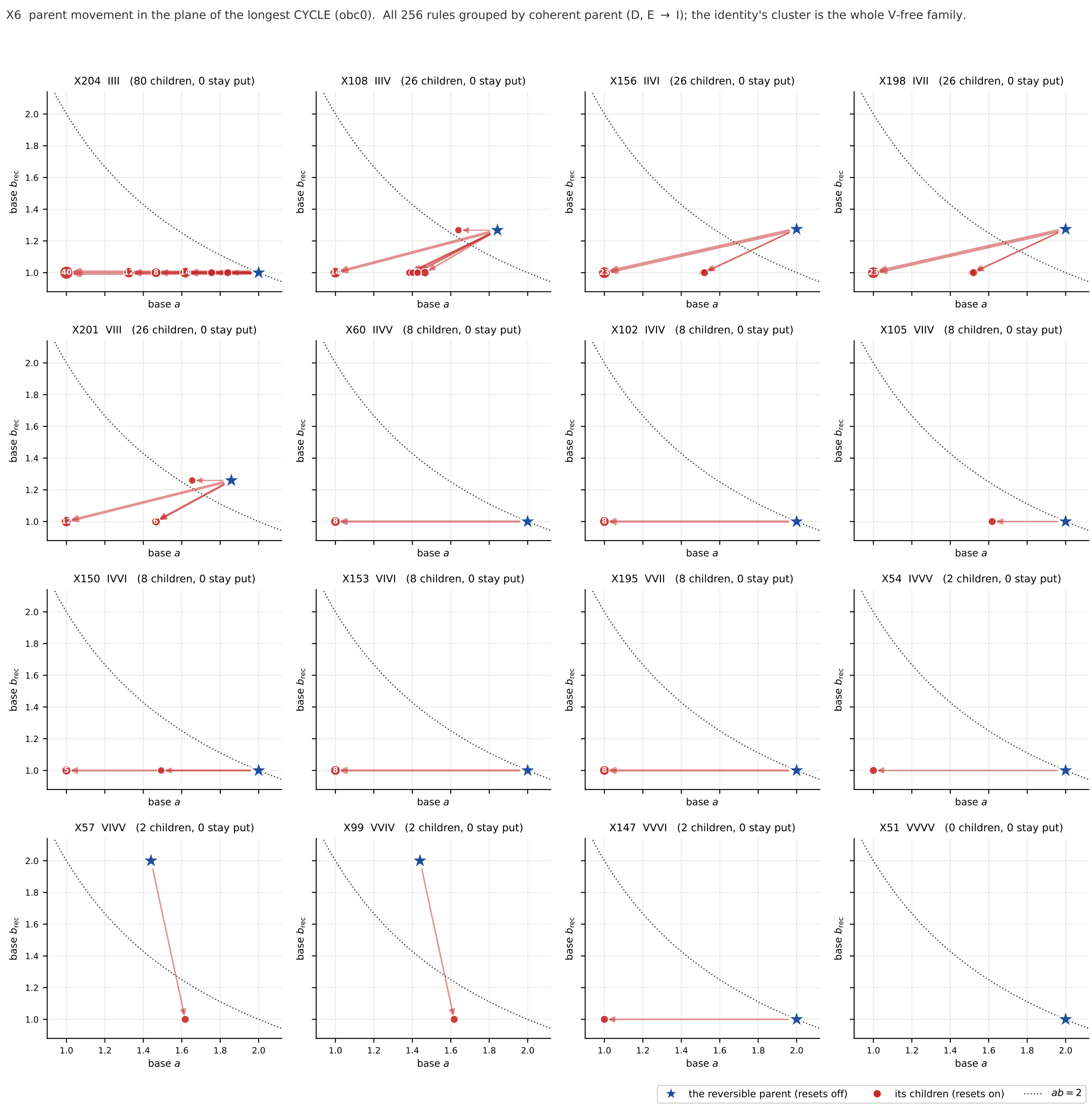

Figure 5: **X6, the same clusters in the cycle plane.** The x-coordinates are identical to X5 by (1); only the heights differ. Thirteen of the fifteen non-empty clusters collapse onto $b_{\text{rec}} = 1$ entirely. Only 108 and 201 keep a child at the parent's height, and those two children are 109 and 73.

holds for all 256 rules at every N, and is tight in the same place — the smallest ratio anywhere is exactly 1.0000, at rule 0, $N = 6$.

Does not carry over. Interference between distinct trajectories of a *basis* state: succ is single-valued, so no two branches can cancel, and at the level of the support the monitored and the unmonitored channel agree. Conversely the transient structure, which was a thin and somewhat awkward part of the Hadamard story, becomes the dominant feature.

What the first draft of this report got wrong. It continued that sentence with "no dark states … the Tier-1d monitored/unmonitored certificate is vacuous here". That is false, and §11 is the correction. A functional graph is a statement about *populations*; a reset is not a unitary, and its two Kraus operators act identically on basis states and completely differently on a coherence. Decoherence-free subspaces exist in this family, are exponentially large, and are invisible to everything above.

The V-free rules are literally the same circuits. A rule with no `V` in its table never invokes the gate, so its Hadamard circuit and its X circuit are the same map, symbol for symbol. That is 81 of the 256 rules (204 plus the 80 V-free ones), and it makes the two stores a genuine cross-implementation test: the Hadamard engine (set-valued successors, union-find over `core/cycle.py`) and the permutation engine (a numpy step table and a three-colour walk) must return identical sector partitions for all 81, and they do, at every N where both were run. More generally the X-gate graph is a *subgraph* of the Hadamard one — flipping the target bit is one of the two Hadamard outcomes — so the X sectors always refine the Hadamard sectors. R11 develops that relation; here it is worth stating because it is the reason the two reports may be compared at all.

The one identity that is a genuine control. $n_{\text{rec}} = n_{\text{wcc}}$ held for **every unit computed**, with zero failures, and the union-find sector decomposition — which knows nothing about functional graphs — reproduced the basin decomposition exactly wherever both were run. Two independent code paths, one theorem, no discrepancies.

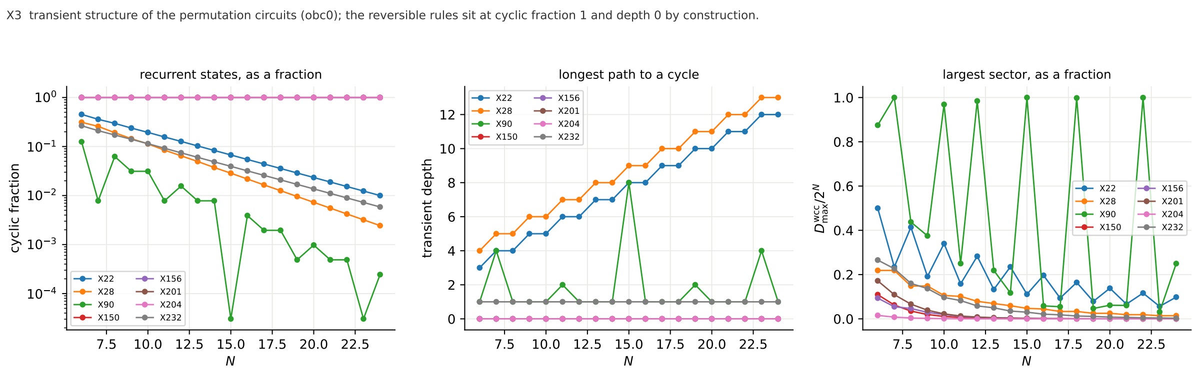

Figure 6: **X3, transient structure.** Cyclic fraction (log scale), transient depth, and largest sector as a fraction of the space, against N. The reversible rules sit at cyclic fraction 1 and depth 0 by construction; the irreversible ones fall away exponentially.

10 Exhibiting the long orbits

§3.1 rests on a fit: eight rules are called exponential because a two-parameter exponential describes their longest-cycle series over $N = 6 \ldots 24$, where the largest number involved is 6.8

million for two of them and 561 for the rest. Fitting an exponential to numbers of order 500 is not a demonstration, so this section attacks the same claim from the other side — by *exhibiting* an orbit and measuring its length. A measured orbit is a lower bound on $L_{\max}$, and a lower bound that keeps growing exponentially is what the claim needs. It is also a proof rather than an estimate: the orbit is there.

Why it is cheap. Following one orbit needs no graph. At `obc0` the even sites are an independent set and so are the odd ones, so the brick-wall sweep is exactly two *simultaneous* sublattice updates — an even site reads only odd bits, which the even half-sweep does not touch. A simultaneous sublattice update is a handful of bitwise operations on the state word:

$$L = (x \ll 1) \wedge \text{mask}, \quad R = x \gg 1, \qquad x \longmapsto \big((x \oplus A_V) \wedge \neg A_D\big) \vee A_E,$$

where A_s collects the sublattice sites whose neighbour pattern selects symbol s. One sweep is about twenty integer operations at any N, and the memory is one machine word — against $O(2^N)$ for the full walk. At 1.2 million sweeps per second this puts $N = 40$ within reach for rules whose orbits are short, and $N = 29$ for rules whose orbit is a finite fraction of the space.

The resets do not break it. The identity is a property of the interaction graph, not of the local action: an even site *writes* an even bit and *reads* only odd bits, so nothing an even site does — `I`, `V`, `D` or `E` — can change an input to another even site. Verified rather than argued: the bitwise sweep agrees with the compiled scalar step (`xca.x_step`) for all 256 rules at $N = 6 \ldots 11$, every state, zero mismatches, and 240 of those rules carry a reset. At `pbc` with odd N the sublattices stop being independent sets and the identity genuinely fails, so the constructor refuses `pbc`.

rule	tuple	exact $L_{\max}$ at $N = 24$	N	orbit exhibited	fraction of 2^N
57	`VIVV`	6 807 384	32	2 078 885 256	0.484
99	`VVIV`	6 807 384	31	986 287 136	0.459
73	`VIID`	561	40	21 505	1.96×10^{-8}
201	`VIII`	561	40	16 269	1.48×10^{-8}
109	`EIIV`	391	40	11 339	1.03×10^{-8}
108	`IIIV`	391	40	8 976	8.16×10^{-9}
156	`IIVI`	420	40	5 460	4.97×10^{-9}
198	`IVII`	420	40	5 460	4.97×10^{-9}

Result. The growth continues, on both scales.

- Rules 57 and 99 have an orbit through a *finite fraction* of the space at every size reached: 31.6 million states at $N = 26$ (47%), 66.9 million at $N = 27$ (50%), 203 million at $N = 29$ (38%), 986 287 136 at $N = 31$ (46%) and 2 078 885 256 at $N = 32$ (48%) — over two billion states in a single orbit. The fit's $b_{\rm rec} = 2$ is not an extrapolation; the orbit is exhibited, and the certified lower bound grows by a factor of 246 over seven sites. ($N = 32$ was run for rule 57 only; the two rules agree exactly at every smaller size.)

- **The fraction genuinely oscillates.** An earlier draft read the dips at $N = 28$ (23%) and $N = 30$ (15%) as two to four draws occasionally missing the largest orbit. Re-running those two sizes with more draws — six at $N = 28$, five at $N = 30$ — reproduced the same maxima exactly (62 326 552 and 159 174 352), so the dips are structure, not undersampling: the largest orbit is a fraction of the space that varies with N rather than a fixed share of it. The certificate is unaffected either way, being a lower bound.

- The other six run from $L_{\max} = 391$–561 at $N = 24$ to 5460–21505 at $N = 40$, a factor of 13–38 over sixteen sites, still on a straight line in Figure 7.

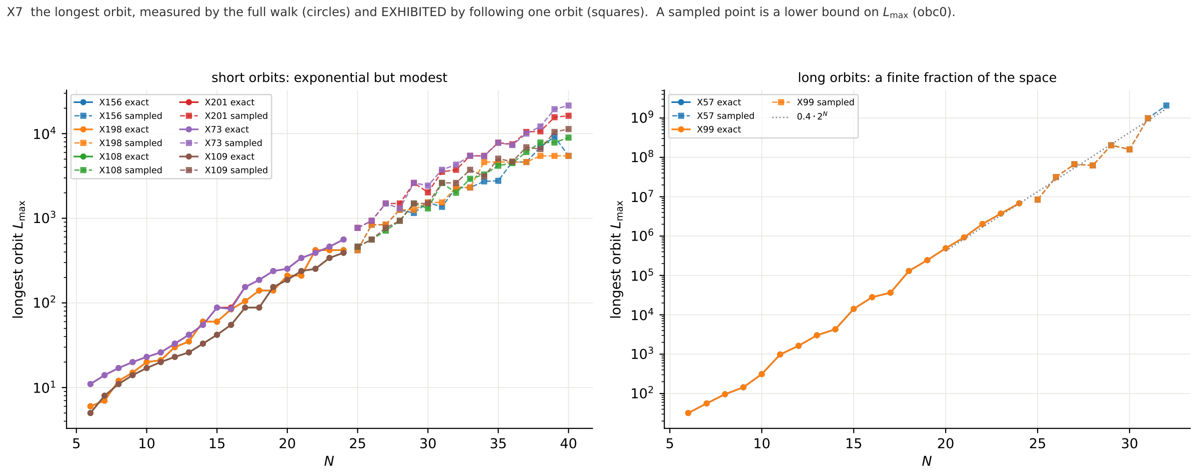

Figure 7: **X7, the longest orbit against** N, log scale, so an exponential is a straight line. Circles: exact, from the full $O(2^N)$ walk. Squares: exhibited, by following a single orbit. Left, the six rules whose orbits are exponential but modest — carried from $N = 24$ to $N = 40$. Right, rules 57 and 99, whose orbit tracks $0.4 \cdot 2^N$ (dotted).

- The lengths stay conspicuously composite: 420, 840, 1260, 1540, 2310, 4620, 6930, 9240 for rule 156, which is the least-common-multiple structure of R11 §7.2 continuing to $N = 40$.

A caveat on the sampled numbers. A sampled point is a lower bound and its tightness degrades with N: the maximal-length sectors are a shrinking share of the space, so a fixed number of draws finds them less often. Fitting the sampled tail gives bases of 1.18–1.23 against the exact fits' 1.26–1.27, and the discrepancy is the bound loosening, not a disagreement about the growth. Read the sampled series as a certificate — there *is* an orbit of at least this length — not as a measurement of $L_{\max}$.

11 Coherence: what the functional graph cannot see

Everything above is about where basis states go. This section is about what happens to a coherence between two of them, and it reverses one of the first draft's conclusions.

11.1 A functional graph does not mean a unitary

A reset is a channel, not a gate. Reset-to-$|0\rangle$ has two Kraus operators, $|0\rangle\langle 0|$ and $|0\rangle\langle 1|$, and *both* send a basis state to one whose target bit is 0 — which is exactly why succ is single-valued and the graph is functional. On a coherence they do something else entirely. For $\rho = |x\rangle\langle y|$ the channel gives $\sum_\mu K_\mu |x\rangle\langle y| K_\mu^\dagger$, and a term survives only if the *same* μ has a non-zero matrix element on both sides. At a reset site that means the coherence survives iff x and y already agree on the target bit when the reset fires. So define the label of one sweep,

$$J(x) \;=\; \{\,\text{sites where a reset fired on a bit not already holding the reset value}\,\},$$

the set of places where the *off*-diagonal Kraus branch was taken. Then

$$|x\rangle\langle y| \;\longmapsto\; \begin{cases} |\Phi(x)\rangle\langle\Phi(y)|, & J(x) = J(y),\\ 0, & \text{otherwise.}\end{cases}$$

J does not affect where a basis state goes, so the functional graph is blind to it. This is the same-label join of the Tier-1d pair graph (R7), specialised to a family in which succ happens to be single-valued.

Proposition 5 (DFS blocks)**.** *$x \sim y$ iff $J(\Phi^t x) = J(\Phi^t y)$ for every $t \geq 0$ is an equivalence relation on the recurrent set, and the span of each class is a decoherence-free subspace: the channel restricted to it is the permutation, hence unitary.*

11.2 §8's inheritance measurement was a DFS criterion

The connection to what has already been computed is exact. "The reset acts as the identity on the recurrent set" — the property §8 verified for 177 of the 240 irreversible rules — says precisely that no reset ever takes its off-diagonal branch there, i.e. $J \equiv \emptyset$ on Rec. Then all labels agree trivially and the *whole recurrent set* is a single DFS. Checked rule by rule: the inheritance flag and the no-jump flag agree for 240 of 240 rules. The measurement was right; the reading of it was missing.

family	rules	DFS of dim > 1	whole Rec one block
reversible	16	16	16
V+reset	160	69	65
V-free	80	58	56
all	256	143	137

At $N = 12$, 127 **of the** 240 **irreversible rules carry a decoherence-free subspace of dimension greater than one**, and for 121 of them it is the entire recurrent set. (The 16 reversible rules trivially have one of dimension 2^N: they are unitary.) So the family has protected coherence almost half the time, and the first draft's "no dark states" was wrong by a wide margin.

11.3 The dimensions are exponential, and they are constrained shifts

rule	tuple	dim DFS, $N = 8, 10, 12, 14, 16$	base	recurrent set
1	`VDDD`	55, 144, 377, 987, 2 584	1.6180	golden-mean shift, no `11`
76	`IIID`	149, 504, 1 705, 5 768, 19 513	1.8393	no `111`
73	`VIID`	149, 504, 1 705, 5 768, 19 513	1.8393	no `111`
205	`EIII`	105, 355, 1 201, 4 063, 13 745	1.8392	tribonacci base, not a no-word shift
109	`EIIV`	105, 355, 1 201, 4 063, 13 745	1.8392	tribonacci base, not a no-word shift
22	`IVVD`	76, 199, 521, 1 364, 3 571	1.6181	Lucas L_{N+1}
232	`DIIE`	44, 117, 305, 798, 2 091	1.6204	base $\to \varphi$, sequence unidentified
90	`DEED`	2, 1, 1, 2, 1	0.9170	labels de-synchronise

Two of these are exactly identifiable, and they are the sort of space the constrained-model literature is about.

- **Rule** 1 (`VDDD`). Its recurrent set — and therefore its DFS — is exactly the golden-mean shift, the strings with no two adjacent 1s, of dimension F_{N+2}. That is the hard-core / Rydberg-blockade constrained space, and here it is a subspace on which a dissipative circuit acts unitarily.

- **Rules** 76 **and** 73 (`IIID`, `VIID`). The strings with no three consecutive 1s, dimension following the tribonacci recurrence $a_N = a_{N-1} + a_{N-2} + a_{N-3}$, base 1.83929, the tribonacci constant.

- **Rule** 22 (`IVVD`) gives the Lucas numbers L_{N+1} exactly: $76, 199, 521, 1364, 3571$ at $N = 8 \dots 16$, base φ.

- **Rule** 232 (`DIIE`) has $44, 117, 305, 798, 2091$, base $\to \varphi$; the sequence is not one we recognise.

And the negative case is equally sharp. Rules 90 and 165 (DEED, EDDE) — the two GF(2)-affine ones of §7 — have long cycles whose reset labels de-synchronise, every block has dimension 1, and no coherence survives at all. R7 reached the same conclusion from the Hadamard side; the criterion is label synchronisation around the cycle, not the mere existence of a cycle.

11.4 Cross-check against the exact channel

Rule 232 is V-free, so it is the *same circuit* under both gates (§9) and the Tier-1b superoperator machinery applies to it unchanged. Its exact Cesaro rank is

$$\dim \mathrm{Fix}(\Phi) \;=\; 49,\; 121,\; 289 \quad \text{at } N = 4, 5, 6 \text{ } (\texttt{obc0}),$$

i.e. n_{rec}^2 with $n_{\mathrm{rec}} = 7, 11, 17$ — a full matrix algebra M_d, which is the signature of a decoherence-free subspace of dimension d. The monitored description sees d classical pointer states and misses all $d^2 - d$ coherences; the Tier-1b classification calls the attractor "$d\times$mixing, $d_k = 1$" because it works class by class and the coherence is *between* classes. Our label census gives the same d from the X side, with no superoperator: 17, 44, 117, 305 at $N = 6, 8, 10, 12$.

11.5 What this does and does not rescue

It does not make the sector structure interesting — a cycle decomposition is still what every bijection has. What it does is locate the quantum content precisely. It is not in where the populations go, which is classical; it is in which coherences the resets are forced to leave alone, and that is a genuinely non-classical, exponentially large structure carried by a circuit whose populations are deterministic. It is also the one place where this family is *not* a control for the Hadamard one: for the 81 V-free rules the channel is identical, so these decoherence-free subspaces are exactly as present in R1–R9 as they are here.

12 Validation

check	applicable	passing	note
sector sizes partition, $\sum_k D_k = 2^N$	4864	4864	asserted in the analysis
$n_{\mathrm{recurrent}} = n_{\mathrm{wcc}}$ (F1)	4864	4864	0 failures
finite-N $n_{\mathrm{wcc}} D_{\max} \geq 2^N$	256	256	tightest 1.0000 at $N = 6$
sector map $ab \geq 2$	249	245	4 below
cycle map $ab \geq 2$ (*not* required)	241	18	223 below, as expected

The last row is the point of the family rather than a failure: the cycle map has no exclusion curve, because cycles do not partition the basis (§5), and 223 of 241 rules do sit below $ab = 2$ there. In the sector map, where the curve does bind, only *four* rules fall below — and none is a violation. All four are the degenerate case R9 §5 documents, where the sector count is sub-exponential so the base product silently drops the polynomial factor, leaving the inequality to degenerate to $b \geq 2$ approached from below:

rule	tuple	n_{wcc} over $N = 19 \dots 24$	ab
25	VIVD	$2, 3, 2, 2, 3, 2$	1.99887
74	DEID	$7, 8, 8, 8, 9, 9$	1.91986
88	DIED	$7, 8, 8, 8, 9, 9$	1.93756
182	IVVE	$11, 11, 12, 12, 13, 13$	1.88127

Two of the four are a cross-check rather than a finding: 74 and 88 are V-free, so by §9 their dynamics is *identical* under both gates, and they are exactly two of the seven rules R9 lists in the same degenerate category. The two families reach the same verdict on the same rules through two independent code paths.

Carrying the sweep from $N = 20$ to $N = 24$ took the count of sub-2 products from 14 to 2 under the fitting convention of the first draft, and to 4 under the corrected one of §3; either way the residue is a finite-size artefact and not a family of offenders.

13 Open items

1. **`pbc`.** Held, as for R9.
2. **Rule 150's cycle length.** $D_{\max} = N + 1$ over $N = 13 \ldots 17$ is five points; the map is GF(2)-linear, so the orbit structure is governed by the order of a matrix over GF(2) and should be derivable rather than fitted. That derivation is not attempted here.
3. **Rules 57 and 99.** A single orbit through a finite fraction of the space — 2.1×10^9 states at $N = 32$ (§10) — while still having exponentially many sectors. Neither the orbit length nor the sector count has been derived, and the derivation now has one more thing to account for: the fraction oscillates with N (confirmed, not a sampling artefact) rather than converging to a constant.
4. **The DFS dimensions.** §11 identifies rule 1's as the golden-mean shift and rules 76/73's as the no-`111` shift, but rule 232's $44, 117, 305, 798, 2091$ and rule 205's tribonacci-base sequence are unidentified, and no general criterion is given for which constrained shift a given table produces.
5. **The recurrent sets that are Fibonacci.** §5 identifies $\mathrm{Rec}(X_1)$ exactly as the golden-mean shift. Twenty-one further rules have $|\mathrm{Rec}| \in \{F_N, F_{N+1}, F_{N+2}\}$ and their sets are *not* the same set; a classification of which constrained shift each one is has not been attempted.
6. **Inheritance as a theorem.** §8 verifies at two sizes that the reset acts as the identity on the recurrent set for 177 of 240 children. A local criterion separating those 177 from the 63 — which reset symbol can fire on a recurrent state, and when — would turn a measurement into a proof.
7. **Comparison with the Hadamard sector map** rule by rule. The two families share a symbol table but not a single quantitative law; whether any rule-level correlation survives the gate change is untested.

14 Reproduce

```
cd code
python -m qca_fragmentation.permutation.sweep --bc obc0 --n-min 6 --n-max 24
python -m qca_fragmentation.permutation.analysis --bc obc0
python -m qca_fragmentation.permutation.movement --bc obc0 \
       --rebuild-inheritance
python -m qca_fragmentation.permutation.tables --bc obc0
python -m qca_fragmentation.permutation.figures --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_xgate.py -q
```

Code: `permutation/xca.py` (the map, ported from `core/cycle.py`), `permutation/functional.py` (the one-pass decomposition and the union-find cross-check), `permutation/analysis.py`, `permutation/movement.py` (the two movement analyses and the inheritance check), `permutation/tables.py`, `permutation/figures.py`.

Data: `results_xgate/{rule}_{bc}.jsonl`, `analytics/xgate_map_obc0.json` and `analytics/xgate_inheritance_obc0.json`.